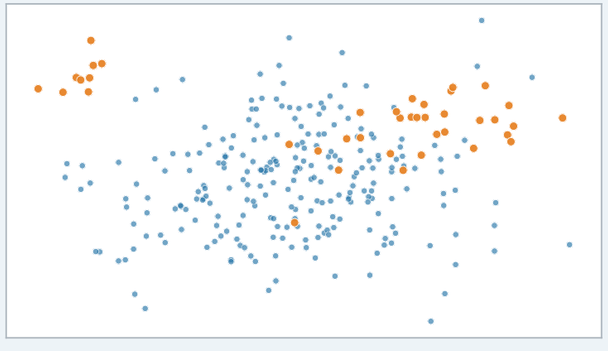
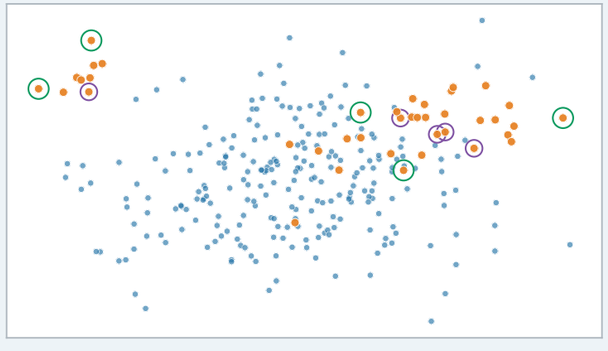
CASB generation: from training data to audited synthetic samples

1 Original training data



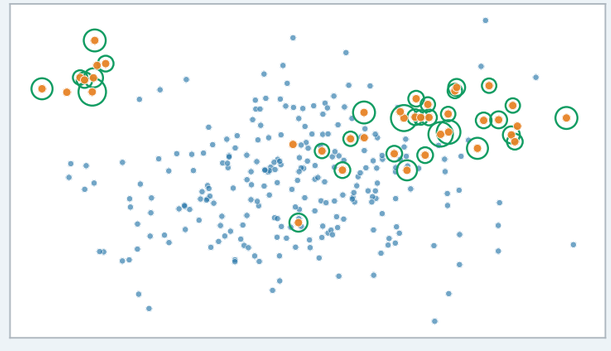
Training-only scaling
240 majority / 40 minority

2 Compute overlapping pools



Coverage: 40 | Border: 5 | Rare: 5
Pools overlap; coverage includes every minority point

3 Select sources within quotas

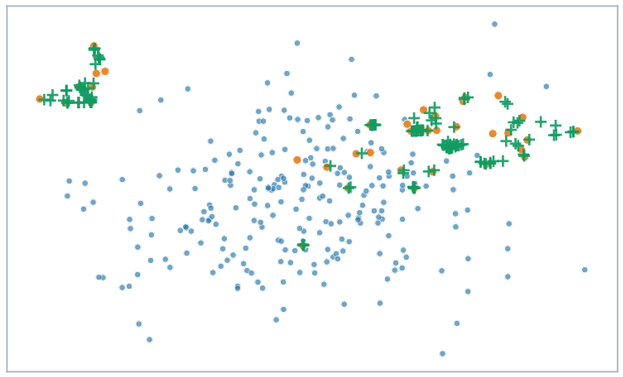


Add 140 samples at ratio 0.75
Branch quotas: 70 / 42 / 28

4 Neighbor + generation rule

Choose t among minority neighbors
z = x_s + lambda (x_t - x_s)

5 Append generated minority points



6 Resampled training set

Base CASB: retain all originals
Edited path: majority-only ENN
Fit SVM, KNN, RF or LR
Evaluate held-out observations

Count-driven generation; geometry is computed on original training observations.
No uncertainty score, minority-noise filtering, or post-generation relocation is performed.